

Epic Messaging — Penetration Test Evidence Pack

Team: kfc

Course: CS4455 Cybersecurity Epic Project 2026

Session date: 2026-06-03

Environment: Production VM — API <https://kfc.theburkenator.com/api/v1>, PostgreSQL secure_messages, Sepolia testnet

Method: Client-driven end-to-end flow per docs/security/penetration_testing_plan.md, with database and worker corroboration.

Executive summary

This document records observed penetration-test evidence from a live demonstration. Test users alice and bob registered, established trust, exchanged end-to-end encrypted messages, forwarded a message, verified blockchain anchors on Sepolia, and demonstrated logout blocking further access.

Database inspection confirms Argon2id password hashes, HMAC-SHA256 refresh-token storage, public E2EE key material only, ciphertext message payloads (no plaintext hello in wire_payload_json), confirmed blockchain anchors with valid 0x transaction hashes, and audit events for auth, keys, messaging, and logout.

Screenshots are sanitised: no passwords, JWTs, refresh tokens, private keys, or seed phrases appear in this pack.

Rubric coverage (summary)

Control area	Evidence sections	Result
Authentication	§1–3, §11, §15–18	Pass — login, Argon2id, refresh HMAC, logout revoke
Access control	§11 (post-logout auth-required)	Pass — session required for /msg
Message confidentiality	§6–9	Pass — ciphertext in DB; hello not found in payload
E2EE / crypto integration	§4–5, §6–8	Pass — public keys, prekeys, encrypted wire payloads
Blockchain integrity	§11–14	Pass — confirmed Sepolia anchors, valid tx hash format
Logging / audit	§16–18	Pass — auth, keys, message, forward, logout events
TLS / HTTPS startup URL	§1 (implicit via successful API calls)	Partial — add dedicated startup/curl screenshot if required
Automated pytest / bandit	Not in screenshot pack	See server/docs/security — run before submission

1. Client registration and login (Alice)

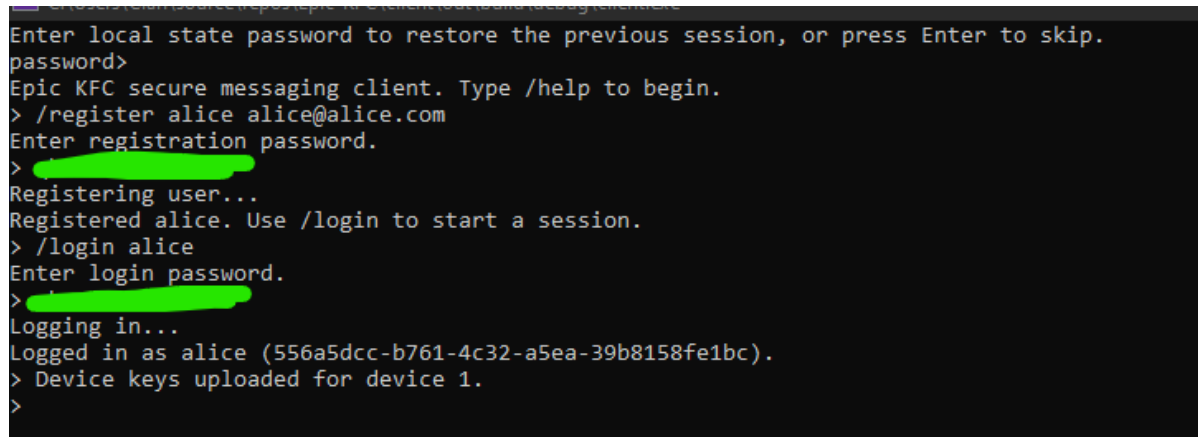
Test users alice and bob were registered and logged in through the Epic secure messaging client against the deployed HTTPS API (kfc.theburkenator.com).

Steps: /register alice, /login alice. Passwords were entered at the prompt and are redacted in the screenshot.

Expected: registration succeeds; login returns a user UUID; device keys upload for device 1.

Observed: Registered alice; logged in as alice (556a5dcc-b761-4c32-a5ea-39b8158fe1bc); Device keys uploaded for device 1.

Rubric: server-side authentication; E2EE key publication.



```
Enter local state password to restore the previous session, or press Enter to skip.
password>
Epic KFC secure messaging client. Type /help to begin.
> /register alice alice@alice.com
Enter registration password.
> [REDACTED]
Registering user...
Registered alice. Use /login to start a session.
> /login alice
Enter login password.
> [REDACTED]
Logging in...
Logged in as alice (556a5dcc-b761-4c32-a5ea-39b8158fe1bc).
> Device keys uploaded for device 1.
>
```

Figure: 01-client-register-login-alice.png

2. Alice and Bob registered — password storage (database)

PostgreSQL on the VM (database `secure_messages`) was inspected after client registration.

Query lists users `alice` and `bob` with `Argon2id` indicator and hash prefix only (not full PHC strings).

Expected: passwords stored as `Argon2id` PHC strings; no plaintext password column.

Observed: both users active; `argon2id_hash = true`; prefixes `$argon2id$v=19$m=65536,...`

Rubric: cryptographic issues (memory-hard password hashing); sensitive data exposure.

```
...
ORDER BY username;
-[ RECORD 1 ]-----+-----
username      | alice
id            | 556a5dcc-b761-4c32-a5ea-39b8158fe1bc
email         | alice@alice.com
is_active     | t
created_at    | 2026-06-03 15:03:36.124954+00
argon2id_hash | t
password_hash_prefix | $argon2id$v=19$m=6
-[ RECORD 2 ]-----+-----
username      | bob
id            | 52c80d80-c879-4dfd-a3ce-8417a2ee86c5
email         | bob@gmail.com
is_active     | t
created_at    | 2026-06-03 15:03:27.757194+00
argon2id_hash | t
password_hash_prefix | $argon2id$v=19$m=6
secure_messages=#
```

Figure: 02-db-users-argon2id.png

3. Login sessions — refresh token storage

Refresh sessions for alice and bob were queried from refresh_sessions joined to users.

Expected: opaque refresh tokens stored only as HMAC-SHA256 digests; ~7-day expiry.

Observed: refresh_hash_prefix values hmac_sha256:...; length 76; revoked_at empty for active sessions.

Rubric: broken authentication; no plaintext refresh tokens in the database.

```
(.venv) student@KFC:~/epic_project/epic-kfc/server/backends$ docker
filter "name=epic-postgres"
docker start epic-postgres
docker port epic-postgres 5432
...
WHERE u.username IN ('alice', 'bob')
ORDER BY rs.created_at DESC
LIMIT 10;
-[ RECORD 1 ]-----+-----
username          | bob
created_at         | 2026-06-03 15:06:20.638772+00
expires_at         | 2026-06-10 15:06:20.73878+00
revoked_at         |
refresh_hash_prefix | hmac_sha256:c5b17f
refresh_hash_length | 76
-[ RECORD 2 ]-----+-----
username          | alice
created_at         | 2026-06-03 15:03:43.656548+00
expires_at         | 2026-06-10 15:03:43.752678+00
revoked_at         |
refresh_hash_prefix | hmac_sha256:00c799
refresh_hash_length | 76
secure_messages=#
```

Figure: 03-db-refresh-tokens-hmac.png

4. Device keys uploaded (public material only)

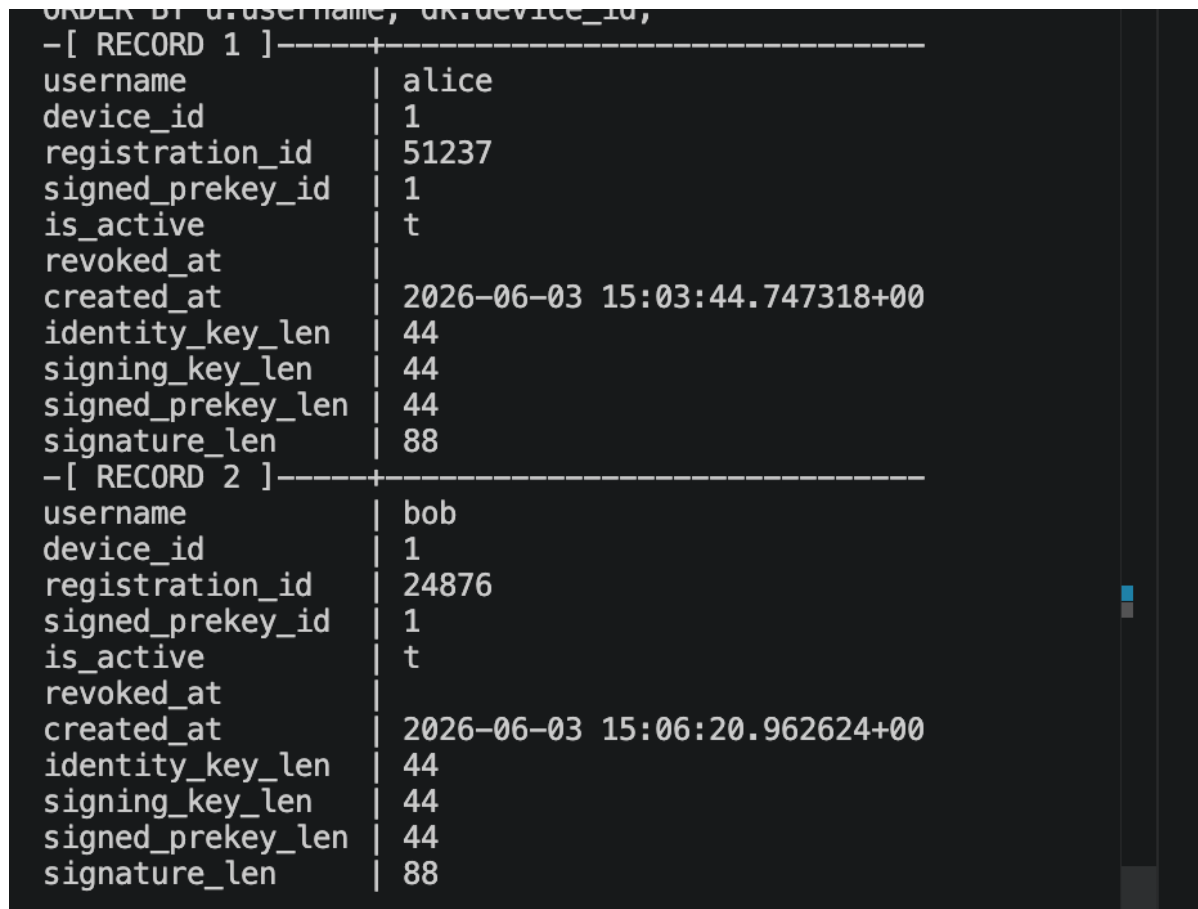
After login, each client uploads identity, signing, and signed pre-key public material.

Query shows key byte lengths only — not private keys.

Expected: public key lengths consistent with base64-encoded 32-byte keys (44 chars) and 64-byte signatures (88 chars).

Observed: alice and bob, device 1, active, identity/signing/signed_prekey length 44, signature 88.

Rubric: E2EE integration; sensitive data exposure (server must not hold private keys).



The image is a screenshot of a terminal window displaying a database query result. The query is ordered by username and device_id. The result shows two records, one for 'alice' and one for 'bob', both with device_id 1. The columns include username, device_id, registration_id, signed_prekey_id, is_active, revoked_at, created_at, identity_key_len, signing_key_len, signed_prekey_len, and signature_len. The key lengths are consistently 44 for identity, signing, and signed_prekey, and 88 for signature.

ORDER BY u.username, dk.device_id,										

-[RECORD 1]-----										
username		alice								
device_id		1								
registration_id		51237								
signed_prekey_id		1								
is_active		t								
revoked_at										
created_at		2026-06-03 15:03:44.747318+00								
identity_key_len		44								
signing_key_len		44								
signed_prekey_len		44								
signature_len		88								
-[RECORD 2]-----										
username		bob								
device_id		1								
registration_id		24876								
signed_prekey_id		1								
is_active		t								
revoked_at										
created_at		2026-06-03 15:06:20.962624+00								
identity_key_len		44								
signing_key_len		44								
signed_prekey_len		44								
signature_len		88								

Figure: 04-db-device-keys-public-material.png

5. One-time prekeys uploaded

Clients upload batches of one-time prekeys for offline session setup (Signal-style directory).

Expected: prekeys recorded per user/device; used count increases after first encrypted send.

Observed at initial capture: 8 prekeys each, all unused (before messaging). After the full E2E flow, prekeys are consumed when bob fetches alice's bundle (see audit log §15).

Rubric: prekey handling; cryptographic session setup.

```
max(opk.created_at) AS last_uploaded,
max(opk.used_at) AS last_used
FROM one_time_prekeys opk
JOIN users u ON u.id = opk.user_id
WHERE u.username IN ('alice', 'bob')
GROUP BY u.username, opk.device_id
ORDER BY u.username, opk.device_id;
```

-[RECORD 1]-----	
username	alice
device_id	1
total_prekeys	8
unused_prekeys	8
used_prekeys	0
first_uploaded	2026-06-03 15:03:46.555659+00
last_uploaded	2026-06-03 15:03:46.555659+00
last_used	
-[RECORD 2]-----	
username	bob
device_id	1
total_prekeys	8
unused_prekeys	8
used_prekeys	0
first_uploaded	2026-06-03 15:06:21.238799+00
last_uploaded	2026-06-03 15:06:21.238799+00
last_used	

Figure: 05-db-one-time-prekeys.png

6–8. Messages between Alice and Bob — ciphertext in database

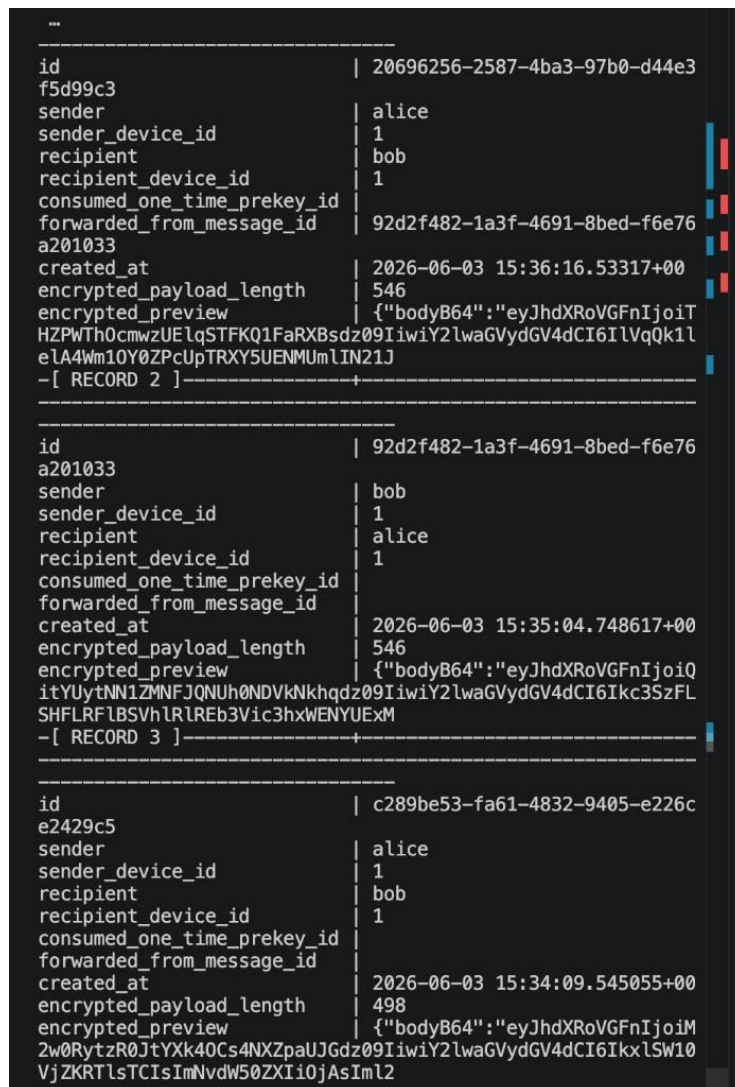
Alice sent hello bob; Bob replied Hello alice, I verified your message; Alice forwarded Bob's message back to Bob with /forward.

Database query on messages shows wire_payload_json / bodyB64 ciphertext previews only.

Expected: no readable message body in database columns.

Observed: encrypted_payload_length 498–546; previews are JSON with bodyB64 and authTag fields; forwarded_from_message_id links alice's forward to bob's original message (92d2f482-1a3f-4691-8bed-f6e76a201033).

Rubric: message confidentiality; sensitive data exposure.



id	20696256-2587-4ba3-97b0-d44e3f5d99c3
sender	alice
sender_device_id	1
recipient	bob
recipient_device_id	1
consumed_one_time_prekey_id	
forwarded_from_message_id	92d2f482-1a3f-4691-8bed-f6e76a201033
created_at	2026-06-03 15:36:16.53317+00
encrypted_payload_length	546
encrypted_preview	{"bodyB64":"eyJhdXRvVGFnIjoiT HZPWTh0cmwzUElqSTFKQ1FaRXBsdz09Iiw iY2lwaGVydGV4dCI6IlVqQk1l eLA4Wm10Y0ZPcUpTRXY5UENMUlIN2Ij -[RECORD 2]-----
id	92d2f482-1a3f-4691-8bed-f6e76a201033
sender	bob
sender_device_id	1
recipient	alice
recipient_device_id	1
consumed_one_time_prekey_id	
forwarded_from_message_id	
created_at	2026-06-03 15:35:04.748617+00
encrypted_payload_length	546
encrypted_preview	{"bodyB64":"eyJhdXRvVGFnIjoIQ itYUytNN1ZMNFJQNUh0NDVhbnkxZDZ0 9IiwiaGVydGV4dCI6Ikc3SzlFL SHFLRF1BSVh1RlREb3Vic3hxWENYUEX M -[RECORD 3]-----
id	c289be53-fa61-4832-9405-e226ce2429c5
sender	alice
sender_device_id	1
recipient	bob
recipient_device_id	1
consumed_one_time_prekey_id	
forwarded_from_message_id	
created_at	2026-06-03 15:34:09.545055+00
encrypted_payload_length	498
encrypted_preview	{"bodyB64":"eyJhdXRvVGFnIjoIM 2w0RytzR0JtYXk4OCs4NXZpaUJGdz09I iwiaGVydGV4dCI6Ikx1SW10 VjZKRTlsTCIsImNvdW50ZXIiOiJAsImL2

Figure: 06-db-messages-encrypted-payloads.png

9. Plaintext not in database (search for message word)

To prove plaintext is not stored, the word hello (from the demo message hello bob) was searched case-insensitively in wire_payload_json for alice↔bob messages.

SQL: m.wire_payload_json ILIKE '%hello%' AS plaintext_found ...

Expected: plaintext_found = false for all rows.

Observed: three message rows; plaintext_found = f for each.

Rubric: message confidentiality; database does not store decryptable plaintext.

```
secure_messages=# SELECT
  m.id,
  su.username AS sender,
  ru.username AS recipient,
  m.wire_payload_json ILIKE '%hello%' AS plaintext_found
FROM messages m
JOIN users su ON su.id = m.sender_user_id
JOIN users ru ON ru.id = m.recipient_user_id
WHERE su.username IN ('alice', 'bob')
      AND ru.username IN ('alice', 'bob')
ORDER BY m.created_at DESC
LIMIT 20;
-[ RECORD 1 ]-----+-----
id              | 20696256-2587-4ba3-97b0-d44e3f5d99c3
sender          | alice
recipient       | bob
plaintext_found | f
-[ RECORD 2 ]-----+-----
id              | 92d2f482-1a3f-4691-8bed-f6e76a201033
sender          | bob
recipient       | alice
plaintext_found | f
-[ RECORD 3 ]-----+-----
id              | c289be53-fa61-4832-9405-e226ce2429c5
sender          | alice
recipient       | bob
plaintext_found | f
```

Figure: 07-db-plaintext-hello-not-found.png

10. Forward evidence (database lineage)

message_forwards table records who forwarded which message to whom, without exposing content.

Expected: alice→bob forward of bob's message to alice appears with correct original_message_id.

Observed: forwarded_message_id 20696256-2587-4ba3-97b0-d44e3f5d99c3; forwarder alice; forwarded_to bob; original_message_id 92d2f482-1a3f-4691-8bed-f6e76a201033.

Rubric: message workflow; access control metadata.

```
ORDER BY f.created_at DESC
LIMIT 10;
```

-[RECORD 1]-----	
forwarded_message_id	20696256-2587-4ba3-97b0-d44e3f5d99c3
created_at	2026-06-03 15:36:16.53317+00
forwarder	alice
forwarded_to	bob
original_message_id	92d2f482-1a3f-4691-8bed-f6e76a201033
original_sender	bob
original_recipient	alice
-[RECORD 2]-----	
forwarded_message_id	16c9cfac-cdb6-462b-bd84-27fe640a6834
created_at	2026-06-03 12:16:53.218885+00
forwarder	cianmc
forwarded_to	testUser
original_message_id	e0bfa6f1-9a84-4c84-a239-a2af9a468a1a
original_sender	testUser
original_recipient	cianmc
-[RECORD 3]-----	
forwarded_message_id	c679d4ff-73d8-4937-95be-86202c9e33c8
created_at	2026-06-03 00:45:30.369544+00
forwarder	cianmc
forwarded_to	danieltest
original_message_id	350e3c1d-ead7-4584-b1d9-b190844e282b
original_sender	danieltest
original_recipient	cianmc
-[RECORD 4]-----	
forwarded_message_id	6a33a946-5073-4e5f-ae0b-e53296cc9a6a
created_at	2026-06-02 22:37:13.35735+00
forwarder	testUser
forwarded_to	cianmc
original_message_id	61a0bf99-cdca-47c9-b02c-b851d4d39193
original_sender	cianmc
original_recipient	testUser

Figure: 08-db-forward-metadata.png

11. Client E2E — send, verify, forward, logout

Two client sessions (alice and bob) plus blockchain worker journal on the VM.

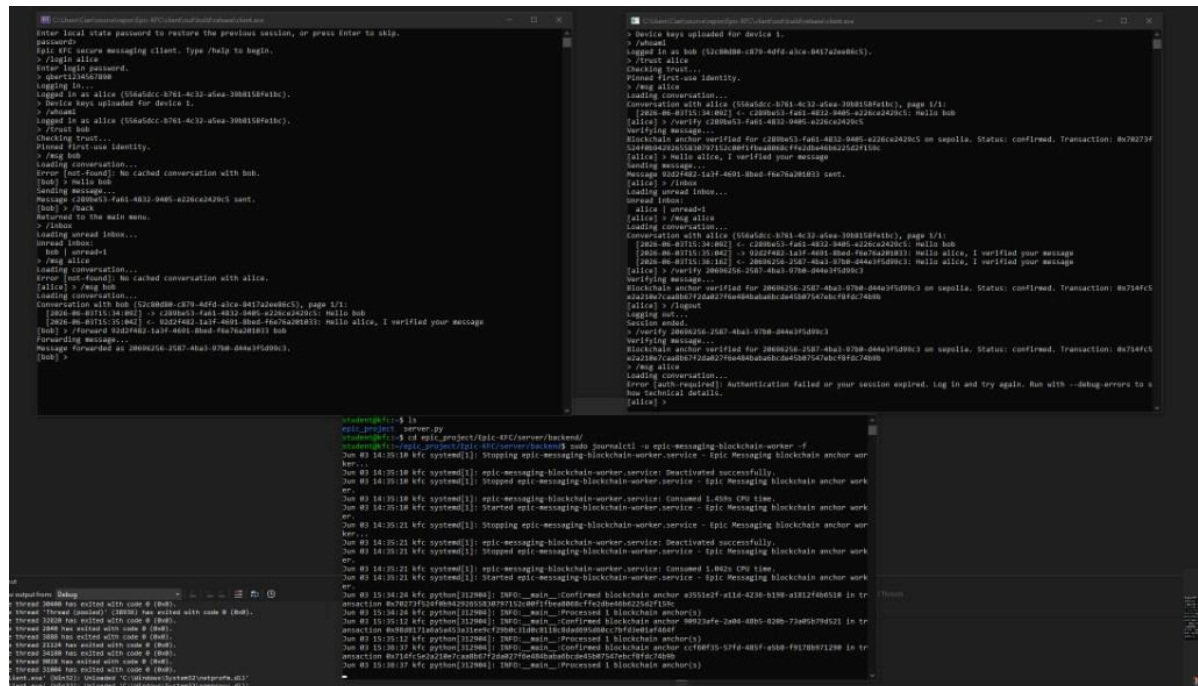
Alice: /trust bob, send hello bob, forward bob's reply to bob.

Bob: /verify on alice's message and on forwarded message — Status confirmed on Sepolia with 0x... hashes.

Bob: /logout then /msg alice → Error [auth-required]: Authentication failed or your session expired.

Worker log: Confirmed blockchain anchor ... transaction 0x70273f... and 0x714fc5... matching client /verify output.

Rubric: functional secure messaging; blockchain fidelity; broken authentication after logout.



```
Enter initial state password to restore the previous session, or press Enter to skip.
Password:
Epic KFC secure messaging (client, type /help to begin).
> /trust alice
Enter login password.
> 0x714fc5f0e8
Logging in...
Logged in as alice (556afdc8-8761-4c32-afca-3088158f61bc).
> /device
Device keys uploaded for device 1.
> /trust bob
Logged in as bob (556afdc8-8761-4c32-afca-3088158f61bc).
> /verify bob
Checking trust...
Forward first user identity.
> /msg bob
Loading conversation...
Error [not-found]: No cached conversation with bob.
[alice] > /msg bob
Sending message...
Message c280b653-fa61-4832-9a85-a226ca242b5c sent.
[alice] > /check
Returned to the main menu.
> /login
Loading saved inbox...
Forward inbox:
[alice] > /msg bob
Loading conversation...
Error [not-found]: No cached conversation with bob.
[alice] > /msg bob
Loading conversation...
Conversation with bob (556afdc8-8761-4c32-afca-3088158f61bc), page 1/1:
[280b-86-8715134-002] < c280b653-fa61-4832-9a85-a226ca242b5c: Hello bob
[280b-86-8715134-002] > 0x2740d3-1a3f-4d81-8ba0-f6a76a2a0883: Hello alice, I verified your message
[alice] > /forward 0x2740d3-1a3f-4d81-8ba0-f6a76a2a0883 bob
Forwarding message...
Message Forwarded as 280b6256-2587-4b41-97b8-0a6a3f5d89c3.
[alice] >
[alice] > /verify 280b6256-2587-4b41-97b8-0a6a3f5d89c3
Verifying message...
Blockchain anchor verified for 280b6256-2587-4b41-97b8-0a6a3f5d89c3 on sepolia, Status: confirmed, Transaction: 0x70273f...
[alice] > /logout
Logging out...
Session ended.
> /verify 280b6256-2587-4b41-97b8-0a6a3f5d89c3
Verifying message...
Blockchain anchor verified for 280b6256-2587-4b41-97b8-0a6a3f5d89c3 on sepolia, Status: confirmed, Transaction: 0x70273f...
[alice] >
[alice] > /msg alice
Loading conversation...
Error [auth-required]: Authentication failed or your session expired. Log in and try again. Run with --debug-errors to see how technical details.
[alice] >

root@kali:~/epic# ./server.py
Epic KFC server.py
[280b6256-2587-4b41-97b8-0a6a3f5d89c3] > epic-messaging-blockchain-worker -f
Jun 03 14:09:18 kfc systemd[1]: Stopping epic-messaging-blockchain-worker.service - Epic Messaging Blockchain anchor worker...
Jun 03 14:09:18 kfc systemd[1]: epic-messaging-blockchain-worker.service: Deactivated successfully.
Jun 03 14:09:18 kfc systemd[1]: Stopped epic-messaging-blockchain-worker.service - Epic Messaging Blockchain anchor worker.
Jun 03 14:09:18 kfc systemd[1]: epic-messaging-blockchain-worker.service: Consumed 1.409s CPU time.
Jun 03 14:09:18 kfc systemd[1]: Started epic-messaging-blockchain-worker.service - Epic Messaging Blockchain anchor worker.
Jun 03 14:09:21 kfc systemd[1]: Stopping epic-messaging-blockchain-worker.service - Epic Messaging Blockchain anchor worker...
Jun 03 14:09:21 kfc systemd[1]: epic-messaging-blockchain-worker.service: Deactivated successfully.
Jun 03 14:09:21 kfc systemd[1]: Stopped epic-messaging-blockchain-worker.service - Epic Messaging Blockchain anchor worker.
Jun 03 14:09:21 kfc systemd[1]: epic-messaging-blockchain-worker.service: Consumed 1.862s CPU time.
Jun 03 14:09:21 kfc systemd[1]: Started epic-messaging-blockchain-worker.service - Epic Messaging Blockchain anchor worker.
Jun 03 15:04:24 kfc python[112084]: INFO: _main...Confirmed blockchain anchor a355a2f7-af1d-423d-9198-a1812f4d0518 in transaction 0x70273f...
Jun 03 15:04:24 kfc python[112084]: INFO: _main...Processed 1 blockchain anchor(s)
Jun 03 15:05:12 kfc python[112084]: INFO: _main...Confirmed blockchain anchor 98b39e-3a8b-48b5-828b-73a0b794521 in transaction 0x70273f...
Jun 03 15:05:12 kfc python[112084]: INFO: _main...Processed 1 blockchain anchor(s)
Jun 03 15:05:37 kfc python[112084]: INFO: _main...Confirmed blockchain anchor c168f35-57f0-485f-a188-f917b8b71288 in transaction 0x714fc5...
Jun 03 15:05:37 kfc python[112084]: INFO: _main...Processed 1 blockchain anchor(s)
Jun 03 15:06:37 kfc python[112084]: INFO: _main...Processed 1 blockchain anchor(s)
```

Figure: 09-client-e2e-verify-forward-logout.png

12. Blockchain anchors — confirmed on Sepolia (detail)

blockchain_anchors rows for alice/bob demo messages after worker confirmation.

Expected: status confirmed; valid 0x + 64 hex transaction_hash; contract on sepolia.

Observed: e.g. message c289be53... tx

0x70273f524f0b94292655830797152c00f1f1bea8068cffe2dbe46b6225d2f159c; contract

0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Faaf; valid_tx_hash true.

Rubric: blockchain integrity; external transaction hash evidence.

<https://sepolia.etherscan.io/tx/0x70273f524f0b94292655830797152c00f1f1bea8068cffe2dbe46b6225d2f159c>

<https://sepolia.etherscan.io/address/0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Faaf>

```

ORDER BY ba.created_at DESC
LIMIT 20;
-[ RECORD 1 ]-----+-----
anchor_id          | ccf60f35-57fd-485f-a5b8-f9178b971290
message_id         | 20696256-2587-4ba3-97b0-d44e3f5d99c3
sender             | alice
recipient          | bob
status             | confirmed
record_id          | 0x8c87eec5f1ad54a4046c83e5cb94da2873ba78
c3a42dacbf6270e387c2f00f22
digest            | 0xdaab89e9472aef19e11ff2100cac461418d63c
2cc24cbfded3b7f506f1d3fd0e
merkle_root        | 0xdaab89e9472aef19e11ff2100cac461418d63c
2cc24cbfded3b7f506f1d3fd0e
transaction_hash   | 0x714fc5e2a210e7caa8b67f2da027f6e484baba
6bcde45b07547ebcf8fdc74b9b
contract_address   | 0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Fa
af
chain              | sepolia
created_at         | 2026-06-03 15:36:16.53317+00
anchored_at        | 2026-06-03 15:36:36+00
valid_tx_hash      | t
-[ RECORD 2 ]-----+-----
anchor_id          | 90923afe-2a04-48b5-820b-73a05b79d521
message_id         | 92d2f482-1a3f-4691-8bed-f6e76a201033
sender             | bob
recipient          | alice
status             | confirmed
record_id          | 0x601dbdad61728a3b59a89e5f87abe04daaee09
a3184b93945df9a45636df743c
digest            | 0xa2c7d6426e56be596e11d658f448819ff34960
990dab866196c0f1abd82ef263
merkle_root        | 0xa2c7d6426e56be596e11d658f448819ff34960
990dab866196c0f1abd82ef263
transaction_hash   | 0x98d8171a6a5a453a31ee9cf29b0c31d0c8118c
8dad695d60cc7bfd3e01af464f
contract_address   | 0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Fa
af
chain              | sepolia
created_at         | 2026-06-03 15:35:04.748617+00
anchored_at        | 2026-06-03 15:35:12+00
valid_tx_hash      | t

```

Figure: 10-db-blockchain-anchors-detail.png

13. Blockchain anchors — summary list

Recent anchors ordered by created_at showing message_id, status, transaction_hash, anchored_at.

Expected: all demo anchors confirmed with Sepolia timestamps on 2026-06-03.

Observed: multiple confirmed rows including forward and reply messages.

Rubric: blockchain worker behaviour; persistence of anchor metadata.

```
LIMIT 10;
-[ RECORD 1 ]-----+-----
id                | ccf60f35-57fd-485f-a5b8-f9178b971290
message_id        | 20696256-2587-4ba3-97b0-d44e3f5d99c3
status            | confirmed
transaction_hash  | 0x714fc5e2a210e7caa8b67f2da027f6e484baba
6bcde45b07547ebcf8fdc74b9b
anchored_at       | 2026-06-03 15:36:36+00
-[ RECORD 2 ]-----+-----
id                | 90923afe-2a04-48b5-820b-73a05b79d521
message_id        | 92d2f482-1a3f-4691-8bed-f6e76a201033
status            | confirmed
transaction_hash  | 0x98d8171a6a5a453a31ee9cf29b0c31d0c8118c
8dad695d60cc7bfd3e01af464f
anchored_at       | 2026-06-03 15:35:12+00
-[ RECORD 3 ]-----+-----
id                | a3551e2f-a11d-4236-b198-a1812f4b6510
message_id        | c289be53-fa61-4832-9405-e226ce2429c5
status            | confirmed
transaction_hash  | 0x70273f524f0b94292655830797152c00f1fbea
8068cffe2dbe46b6225d2f159c
anchored_at       | 2026-06-03 15:34:24+00
-[ RECORD 4 ]-----+-----
id                | b550b2ae-f3af-4e79-9903-17600acc91bf
message_id        | b21b0e61-ba37-48d7-a0a3-b672ad626b73
status            | confirmed
transaction_hash  | 0xb7e031a1356521e510fa3bec224ac0141cd18a
7d0ec20e9c53886e4562fba422
anchored_at       | 2026-06-03 14:27:12+00
-[ RECORD 5 ]-----+-----
id                | c11eef9d-d17c-4ede-b9e6-08eed60d968a
message_id        | 16c9cfac-cdb6-462b-bd84-27fe640a6834
status            | confirmed
transaction_hash  | 0xd586c69b8440ab3dc4244b31191770a2b41535
2aec03a1764c3bf3269801efa7
anchored_at       | 2026-06-03 12:17:12+00
-[ RECORD 6 ]-----+-----
id                | 36554230-1f97-475a-a7a6-f7c6ccaf24c3
```

Figure: 11-db-blockchain-anchors-list.png

14. Blockchain transaction hash validity

Validation query: transaction_hash must match ^0x[0-9a-fA-F]{64}\$.

Expected: no invalid rows in recent anchors.

Observed: valid_tx_hash = t for displayed records; Sepolia chain; anchored_at populated.

Rubric: improper input validation; blockchain metadata quality.

```
transaction_hash ~ '^0x[0-9a-fA-F]{64}$' AS valid_tx_hash
```

FROM blockchain_anchors	
ORDER BY created_at DESC	
LIMIT 20;	
-[RECORD 1]-----	
id	ccf60f35-57fd-485f-a5b8-f9178b971290
message_id	20696256-2587-4ba3-97b0-d44e3f5d99c3
status	confirmed
transaction_hash	0x714fc5e2a210e7caa8b67f2da027f6e484baba6bcde45b07547ebcf8fdc74b9b
contract_address	0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Faaf
chain	sepolia
anchored_at	2026-06-03 15:36:36+00
valid_tx_hash	t
-[RECORD 2]-----	
id	90923afe-2a04-48b5-820b-73a05b79d521
message_id	92d2f482-1a3f-4691-8bed-f6e76a201033
status	confirmed
transaction_hash	0x98d8171a6a5a453a31ee9cf29b0c31d0c8118c8dad695d60cc7bfd3e01af464f
contract_address	0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Faaf
chain	sepolia
anchored_at	2026-06-03 15:35:12+00
valid_tx_hash	t
-[RECORD 3]-----	
id	a3551e2f-a11d-4236-b198-a1812f4b6510
message_id	c289be53-fa61-4832-9405-e226ce2429c5
status	confirmed
transaction_hash	0x70273f524f0b94292655830797152c00f1fbea8068cffe2dbe46b6225d2f159c
contract_address	0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Faaf
chain	sepolia
anchored_at	2026-06-03 15:34:24+00
valid_tx_hash	t
-[RECORD 4]-----	
id	b550b2ae-f3af-4e79-9903-17600acc91bf
message_id	b21b0e61-ba37-48d7-a0a3-b672ad626b73
status	confirmed
transaction_hash	0xb7e031a1356521e510fa3bec224ac0141cd18a7d0ec20e9c53886e4562fba422
contract_address	0xcB3EE0E96aCC28c1C606c7598Cf859E8FAD5Faaf
chain	sepolia
anchored_at	2026-06-03 14:27:12+00
valid_tx_hash	t

Figure: 12-db-blockchain-tx-hash-validity.png

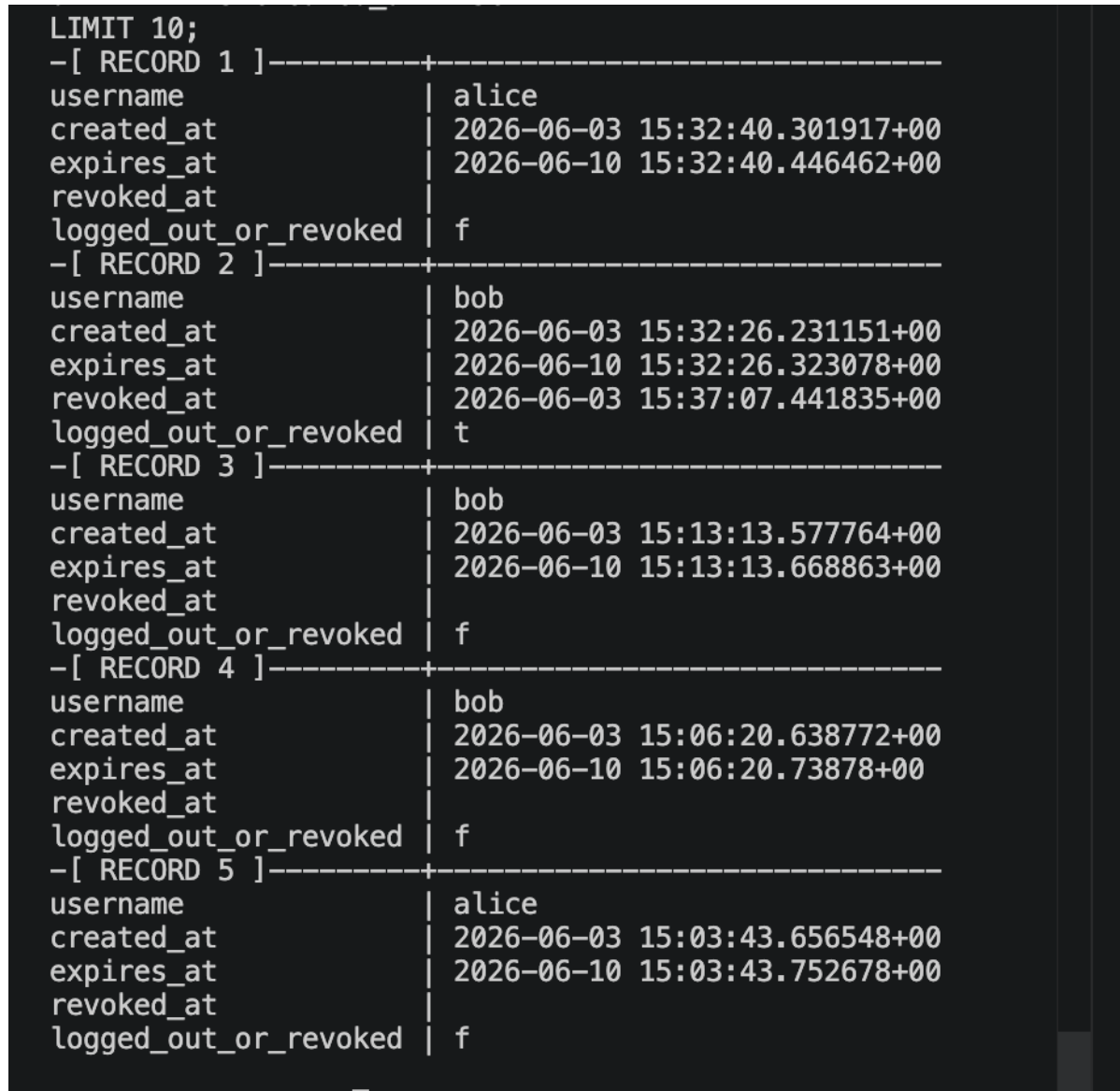
15. Logout — refresh session revoked

After bob /logout, refresh_sessions shows revoked_at set for the active session.

Expected: logged_out_or_revoked = true for ended session; new commands require re-login.

Observed: bob session created 15:32:26 revoked at 15:37:07; logged_out_or_revoked = t.

Rubric: session termination; broken authentication (negative test after logout).



```
LIMIT 10;
-[ RECORD 1 ]-----+-----
username          | alice
created_at        | 2026-06-03 15:32:40.301917+00
expires_at        | 2026-06-10 15:32:40.446462+00
revoked_at        |
logged_out_or_revoked | f
-[ RECORD 2 ]-----+-----
username          | bob
created_at        | 2026-06-03 15:32:26.231151+00
expires_at        | 2026-06-10 15:32:26.323078+00
revoked_at        | 2026-06-03 15:37:07.441835+00
logged_out_or_revoked | t
-[ RECORD 3 ]-----+-----
username          | bob
created_at        | 2026-06-03 15:13:13.577764+00
expires_at        | 2026-06-10 15:13:13.668863+00
revoked_at        |
logged_out_or_revoked | f
-[ RECORD 4 ]-----+-----
username          | bob
created_at        | 2026-06-03 15:06:20.638772+00
expires_at        | 2026-06-10 15:06:20.73878+00
revoked_at        |
logged_out_or_revoked | f
-[ RECORD 5 ]-----+-----
username          | alice
created_at        | 2026-06-03 15:03:43.656548+00
expires_at        | 2026-06-10 15:03:43.752678+00
revoked_at        |
logged_out_or_revoked | f
```

-[RECORD 1]-----+-----	
username	alice
created_at	2026-06-03 15:32:40.301917+00
expires_at	2026-06-10 15:32:40.446462+00
revoked_at	
logged_out_or_revoked	f
-[RECORD 2]-----+-----	
username	bob
created_at	2026-06-03 15:32:26.231151+00
expires_at	2026-06-10 15:32:26.323078+00
revoked_at	2026-06-03 15:37:07.441835+00
logged_out_or_revoked	t
-[RECORD 3]-----+-----	
username	bob
created_at	2026-06-03 15:13:13.577764+00
expires_at	2026-06-10 15:13:13.668863+00
revoked_at	
logged_out_or_revoked	f
-[RECORD 4]-----+-----	
username	bob
created_at	2026-06-03 15:06:20.638772+00
expires_at	2026-06-10 15:06:20.73878+00
revoked_at	
logged_out_or_revoked	f
-[RECORD 5]-----+-----	
username	alice
created_at	2026-06-03 15:03:43.656548+00
expires_at	2026-06-10 15:03:43.752678+00
revoked_at	
logged_out_or_revoked	f

Figure: 13-db-sessions-revoked-on-logout.png

16. Audit timeline — messaging and forward

audit_log entries for the demo session (newest first).

Observed: message.sent (alice, bob), message.forwarded (alice), keys.prekey_bundle_fetched, auth.logout.

Rubric: security logging; penetration-test audit trail.

```
LIMIT 40;
-[ RECORD 1 ]-----
created_at | 2026-06-03 15:37:07.445205+00
actor      |
event_type | auth.logout
success    | t
resource_type |
resource_id |
details    | null
-[ RECORD 2 ]-----
created_at | 2026-06-03 15:36:16.544573+00
actor      | alice
event_type | message.forwarded
success    | t
resource_type | message
resource_id | 20696256-2587-4ba3-97b0-d44e3f5d99c3
details    | {"forwarded_from_message_id": "92d2f482-1a3f-4691-8bed-f6e76a201033"}
-[ RECORD 3 ]-----
created_at | 2026-06-03 15:35:04.768473+00
actor      | bob
event_type | message.sent
success    | t
resource_type | message
resource_id | 92d2f482-1a3f-4691-8bed-f6e76a201033
details    | null
-[ RECORD 4 ]-----
created_at | 2026-06-03 15:34:09.582354+00
actor      | alice
event_type | message.sent
success    | t
resource_type | message
resource_id | c289be53-fa61-4832-9405-e226ce2429c5
details    | null
-[ RECORD 5 ]-----
created_at | 2026-06-03 15:33:30.107587+00
actor      | bob
event_type | keys.prekey_bundle_fetched
success    | t
resource_type | device_key
resource_id | 5ad72e26-45d2-4f19-879e-c07c1225e016
details    | {"target_device_id": 1, "one_time_prekey_included": true}
-[ RECORD 6 ]-----
created_at | 2026-06-03 15:33:19.517216+00
actor      | alice
event_type | keys.prekey_bundle_fetched
```

Figure: 14-db-audit-messaging-forward.png

17. Audit timeline — authentication and keys

Earlier audit entries: auth.login_success for alice and bob, keys.device_upserted, keys.prekey_bundle_fetched.

Rubric: authentication monitoring; key-directory events without secrets in details.

resource_id	5ad72e26-45d2-4f19-879e-c07c1225e016
details	{ "target_device_id": 1, "one_time_prekey_included": true }
- [RECORD 6] ->	
created_at	2026-06-03 15:33:19.517216+00
actor	alice
event_type	keys.prekey_bundle_fetched
success	t
resource_type	device_key
resource_id	72654a61-c66a-477d-a22f-ee1e0c60130c
details	{ "target_device_id": 1, "one_time_prekey_included": true }
- [RECORD 7] ->	
created_at	2026-06-03 15:32:45.714832+00
actor	alice
event_type	keys.device_upserted
success	t
resource_type	device_key
resource_id	5ad72e26-45d2-4f19-879e-c07c1225e016
details	{ "device_id": 1 }
- [RECORD 8] ->	
created_at	2026-06-03 15:32:40.450493+00
actor	alice
event_type	auth.login_success
success	t
resource_type	user
resource_id	556a5dcc-b761-4c32-a5ea-39b8158fe1bc
details	null
- [RECORD 9] ->	
created_at	2026-06-03 15:32:31.674439+00
actor	bob
event_type	keys.device_upserted
success	t
resource_type	device_key
resource_id	72654a61-c66a-477d-a22f-ee1e0c60130c
details	{ "device_id": 1 }
- [RECORD 10] ->	
created_at	2026-06-03 15:32:26.327179+00
actor	bob
event_type	auth.login_success
success	t
resource_type	user
resource_id	52c80d80-c879-4dfd-a3ce-8417a2ee86c5
details	null
- [RECORD 11] ->	
created_at	2026-06-03 15:13:14.660081+00
actor	bob

Figure: 15-db-audit-auth-and-keys.png

18. Failed login and logout (audit)

Historical audit entries show auth.login_failed with reason invalid_credentials and successful auth.logout.

Supports negative testing: wrong passwords do not create sessions.

Rubric: broken authentication (failed login); secure failure handling.

```
resource_type |
resource_id   |
details       | null
-[ RECORD 15 ]+-----
created_at    | 2026-06-03 00:29:33.136464+00
actor         |
event_type    | auth.login_failed
success       | f
resource_type |
resource_id   |
details       | {"reason": "invalid_credentials"}
-[ RECORD 16 ]+-----
created_at    | 2026-06-03 00:12:02.143326+00
actor         |
event_type    | auth.logout
success       | t
resource_type |
resource_id   |
details       | null
-[ RECORD 17 ]+-----
created_at    | 2026-06-02 22:34:36.57468+00
actor         |
event_type    | auth.logout
success       | t
resource_type |
resource_id   |
details       | null
-[ RECORD 18 ]+-----
created_at    | 2026-06-02 22:33:28.788308+00
actor         |
event_type    | auth.logout
success       | t
resource_type |
resource_id   |
details       | null
-[ RECORD 19 ]+-----
created_at    | 2026-06-02 22:31:42.628602+00
actor         |
event_type    | auth.login_failed
success       | f
resource_type |
resource_id   |
details       | {"reason": "invalid_credentials"}
-[ RECORD 20 ]+-----
created_at    | 2026-06-02 22:31:26.360636+00
actor         |
event_type    | auth.login_failed
success       | f
resource_type |
resource_id   |
details       | {"reason": "invalid_credentials"}

secure_messages=#
```

Figure: 16-db-audit-login-failed-logout.png